

# JEE MAIN REVISION MASTER: COMMUNICATION SYSTEMS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                              | Frequency | Category         |
|---------------|-------------------------------------------|-----------|------------------|
| Standard      | EM Wave Propagation & Space Communication | 2 PYQs    | Wave Propagation |
| Standard      | Amplitude Modulation & Sideband Analysis  | 0 PYQs    | Modulation       |

## 2. Master Formula Reference Card

| Concept / Topic                           | Mathematical Formulation & Governing Equations                                                                                                                                     |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EM Wave Propagation & Space Communication | $d_{\text{max}} = \sqrt{2Rh_T} + \sqrt{2Rh_R}, \quad f_c = 9\sqrt{N_{\text{max}}},$ $f_{\text{MUF}} = f_c \sec \theta_i, \quad \text{Antenna length} \geq \frac{\lambda}{4}$       |
| Amplitude Modulation & Sideband Analysis  | $\mu = \frac{A_m}{A_c} = \frac{A_{\text{max}} - A_{\text{min}}}{A_{\text{max}} + A_{\text{min}}}, \quad \text{Bandwidth} = 2f_m, \quad P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 EM Wave Propagation & Space Communication (Standard Yield — 2 PYQs)

**Core Principle:** Propagation modes of electromagnetic waves (ground, sky, and space waves), critical frequency, MUF, and antenna LOS range calculations.

■ **Caution / JEE Pitfall:** Ensure earth radius  $R \approx 6400 \text{ km}$  and heights  $h_T, h_R$  are converted into the exact same units before square rooting.

■ **Speed Trick:** For an antenna of height  $h$ , maximum range is  $d = \sqrt{2Rh}$ , and population covered is  $\pi d^2 \times \rho$ .

**Representative Source PYQs:**

- [2018 Shift] Q#1 (MCQ): "The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is..." [Ans: N/A]
- [2017 Shift] Q#2 (MCQ): "A basic communication system consists of (i) Transmitter (ii) Information sources (iii) User of information (iv) Channel (v) Receiver..." [Ans: N/A]

### 3.2 Amplitude Modulation & Sideband Analysis (Standard Yield — 0 PYQs)

**Core Principle:** Principles of amplitude modulation, carrier and sideband components ( $f_c \pm f_m$ ), modulation index, and power relationships.

■ ■ **Caution / JEE Pitfall:** Modulation index  $\mu$  must satisfy  $\mu \leq 1$  (100%) to prevent envelope distortion.

■ **Speed Trick:** Carrier power does not depend on modulation index  $\mu$ , only sideband power increases as  $\mu$  increases.

**Representative Source PYQs:**

• [2018 Shift] Q#1 (MCQ): "The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is..."

[Ans: N/A]

• [2017 Shift] Q#2 (MCQ): "A basic communication system consists of (i) Transmitter (ii) Information sources (iii) User of information (iv) Channel (v) Receiver..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2018 (Q#1)

The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is

- (A) 800 km
- (B) 100 km
- (C) 80 km
- (D) 10 km

**Correct Answer: None**

**Solution Analysis:** To find the range of the antenna (i.e., the distance to the horizon), we can use the formula:  $d = \sqrt{2R\cdot}$  where:  $\bullet$  R is the radius of the Earth  $\bullet$   $\cdot$  is the height of the antenna Since the Earth's radius is given as 6400 km and the antenna's height is 500 m, we first convert the height into kilometers:  $\bullet$  500m=0.5km Now, plug the values into the formula:  $d = \sqrt{2 \times 6400 \times 0.5} = \sqrt{6400} = 80\text{km}$  Thus, the range of the antenna is 80 km, which corresponds to Option C.